

CHECK WHAT YOU LEARNT

Check Yourself - Unit 2**A. Choose the correct answer**

- The actual computer on an Arduino Uno is
(a) the USB socket (b) the ATmega328P chip (c) the blue board (d) the crystal
- The 16 MHz crystal provides
(a) power (b) the clock (c) memory (d) the reset
- Your compiled program is stored in
(a) SRAM (b) flash memory (c) EEPROM (d) the ADC
- The ATmega's ADC is 10-bit, so it reports
(a) 10 values (b) 256 values (c) 1024 values (d) 1023 volts
- One ADC step across a 5V range is about
(a) 4.9 millivolts (b) 49 millivolts (c) 0.5 volts (d) 5 volts
- The bootloader's job is to
(a) run your program (b) receive your code and write it to flash (c) measure voltage (d) power the board

B. Fill in the blanks

- Variables live in _____ while the program is running.
- A 10-bit ADC reports numbers from _____ to _____.
- The smallest change a system can detect is called its _____.
- Ready-made code written by somebody else is called a _____.

C. Answer in one or two lines

- Describe the four stages between pressing Upload and your program running.
- A pin sits at 3.75 volts. Roughly what will `analogRead()` return? Show your working.
- Explain why a program with many long text messages may run out of memory.
- Why is using a library not cheating?

UNIT 3

Basic Project: Fire Security Alarm System

Sessions 6, 7 and 8

Your first machine has a serious job. It must notice a fire quickly, be certain enough not to cry wolf, and once it has decided, refuse to go quiet until a human says so. Every one of those requirements changes the code.

By the end of this unit you will be able to:

- Explain how a flame sensor detects fire without measuring heat.
- Combine two sensors so that both must agree before an alarm sounds.
- Implement a latching alarm with a manual reset.
- Test honestly for false positives and false negatives.

SESSION 6

one class